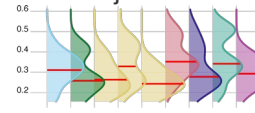
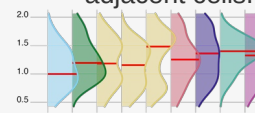
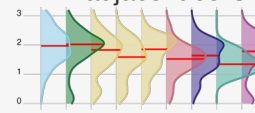
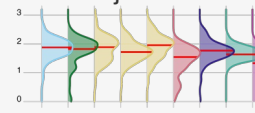
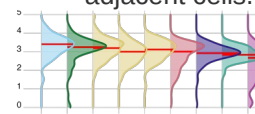
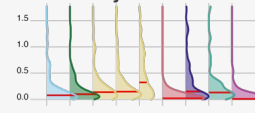
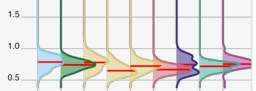
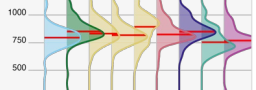
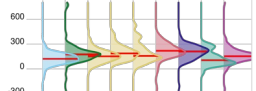
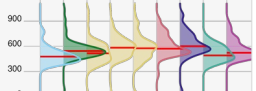
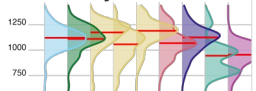


Table S2. Full near-eye metric dictionary and distributions

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Duration													
Time above 1,000 lx meEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) <i>01:10 ± 01:28</i> N=141; d=816	01:18 (00:31, 03:24) <i>02:00 ± 01:54</i> N=13; d=78	01:18 (00:29, 02:12) <i>01:33 ± 01:28</i> N=13; d=78	01:03 (00:20, 02:32) <i>01:48 ± 02:14</i> N=18; d=107	00:40 (00:07, 01:25) <i>01:06 ± 01:22</i> N=26; d=150	00:39 (00:10, 01:46) <i>01:09 ± 01:18</i> N=10; d=60	00:27 (00:06, 00:53) <i>00:38 ± 00:47</i> N=23; d=129	00:30 (00:06, 01:04) <i>00:44 ± 00:51</i> N=17; d=101	00:19 (00:09, 00:43) <i>00:32 ± 00:33</i> N=6; d=32	00:43 (00:17, 01:14) <i>00:58 ± 00:57</i> N=15; d=81	Symlog (base 10; threshold 1)	Time above 1,000 lx meEDI distribution by site. Exact numerical summaries are in the adjacent cells.
Time above 250 lx meEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) <i>03:00 ± 02:29</i> N=141; d=737	04:06 (02:30, 06:15) <i>04:21 ± 02:24</i> N=13; d=73	03:29 (02:03, 05:16) <i>03:39 ± 02:25</i> N=13; d=64	03:42 (02:05, 05:58) <i>04:05 ± 02:51</i> N=18; d=90	02:01 (00:47, 04:20) <i>02:52 ± 02:39</i> N=26; d=141	02:38 (00:55, 04:19) <i>02:57 ± 02:19</i> N=10; d=54	02:43 (00:54, 04:43) <i>03:02 ± 02:19</i> N=23; d=113	01:30 (00:41, 03:26) <i>02:09 ± 01:57</i> N=17; d=94	01:39 (00:33, 02:40) <i>01:48 ± 01:21</i> N=6; d=30	01:14 (00:25, 02:37) <i>01:45 ± 01:46</i> N=15; d=78	Symlog (base 10; threshold 1)	Time above 250 lx meEDI during wake distribution by site. Exact numerical summaries are in the adjacent cells.
Time below 10 lx meEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) <i>01:51 ± 01:02</i> N=139; d=655	02:16 (01:30, 02:39) <i>02:05 ± 00:53</i> N=13; d=67	01:50 (01:21, 02:52) <i>01:59 ± 00:57</i> N=12; d=50	01:42 (00:53, 02:17) <i>01:41 ± 00:58</i> N=18; d=84	01:52 (01:04, 02:25) <i>01:52 ± 01:06</i> N=26; d=132	01:43 (00:42, 02:32) <i>01:45 ± 01:10</i> N=10; d=52	01:43 (00:58, 02:50) <i>01:50 ± 01:05</i> N=23; d=105	01:31 (00:49, 02:15) <i>01:38 ± 01:00</i> N=17; d=82	02:16 (01:15, 02:39) <i>01:56 ± 01:00</i> N=6; d=25	02:30 (01:28, 02:52) <i>02:09 ± 00:57</i> N=14; d=58	Symlog (base 10; threshold 1)	Time below 10 lx meEDI before sleep distribution by site. Exact numerical summaries are in the adjacent cells.
Time below 1 lx meEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) <i>07:04 ± 01:57</i> N=141; d=778	07:08 (06:03, 08:18) <i>07:12 ± 01:46</i> N=13; d=76	07:14 (05:56, 08:06) <i>07:01 ± 01:50</i> N=13; d=68	07:08 (05:56, 08:16) <i>07:05 ± 01:50</i> N=18; d=95	06:54 (05:51, 07:46) <i>06:47 ± 01:30</i> N=26; d=150	05:52 (04:41, 07:03) <i>05:58 ± 01:58</i> N=10; d=60	08:06 (07:10, 09:10) <i>08:02 ± 01:57</i> N=23; d=118	06:45 (05:08, 08:15) <i>06:40 ± 02:07</i> N=17; d=99	06:56 (06:06, 07:36) <i>06:39 ± 01:43</i> N=6; d=31	07:31 (06:31, 09:00) <i>07:35 ± 02:15</i> N=15; d=81	Symlog (base 10; threshold 1)	Time below 1 lx meEDI during sleep distribution by site. Exact numerical summaries are in the adjacent cells.
Longest period above 250 lx meEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) <i>00:55 ± 01:00</i> N=141; d=816	00:57 (00:28, 01:27) <i>01:12 ± 01:02</i> N=13; d=78	00:48 (00:23, 01:20) <i>01:09 ± 01:15</i> N=13; d=78	01:00 (00:21, 01:40) <i>01:20 ± 01:31</i> N=18; d=107	00:35 (00:13, 01:06) <i>00:50 ± 00:54</i> N=26; d=150	00:38 (00:17, 01:32) <i>01:01 ± 01:02</i> N=10; d=60	00:38 (00:16, 01:04) <i>00:45 ± 00:36</i> N=23; d=129	00:34 (00:14, 00:55) <i>00:43 ± 00:41</i> N=17; d=101	00:28 (00:16, 00:43) <i>00:30 ± 00:20</i> N=6; d=32	00:23 (00:15, 00:49) <i>00:38 ± 00:45</i> N=15; d=81	Symlog (base 10; threshold 1)	Longest period above 250 lx meEDI distribution by site. Exact numerical summaries are in the adjacent cells.
Dynamics													

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) <i>0.318 ± 0.095</i> N=141; d=816	0.311 (0.272, 0.367) <i>0.315 ± 0.078</i> N=13; d=78	0.258 (0.239, 0.389) <i>0.288 ± 0.094</i> N=13; d=78	0.264 (0.219, 0.319) <i>0.274 ± 0.074</i> N=18; d=107	0.329 (0.263, 0.424) <i>0.344 ± 0.107</i> N=26; d=150	0.244 (0.218, 0.277) <i>0.252 ± 0.058</i> N=10; d=60	0.352 (0.321, 0.455) <i>0.368 ± 0.092</i> N=23; d=129	0.278 (0.254, 0.394) <i>0.323 ± 0.104</i> N=17; d=101	0.342 (0.315, 0.473) <i>0.37 ± 0.118</i> N=6; d=32	0.293 (0.255, 0.354) <i>0.299 ± 0.062</i> N=15; d=81	Identity	Interdaily stability distribution by site. Exact numerical summaries are in the adjacent cells. 
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) <i>1.229 ± 0.389</i> N=141; d=816	0.997 (0.818, 1.388) <i>1.088 ± 0.343</i> N=13; d=78	1.193 (0.96, 1.389) <i>1.238 ± 0.345</i> N=13; d=78	1.182 (0.928, 1.49) <i>1.177 ± 0.407</i> N=18; d=107	1.154 (0.801, 1.442) <i>1.125 ± 0.354</i> N=26; d=150	1.48 (0.869, 1.642) <i>1.332 ± 0.477</i> N=10; d=60	1.251 (0.999, 1.497) <i>1.251 ± 0.413</i> N=23; d=129	1.359 (0.95, 1.546) <i>1.307 ± 0.415</i> N=17; d=101	1.399 (1.282, 1.483) <i>1.355 ± 0.256</i> N=6; d=32	1.324 (1.029, 1.595) <i>1.344 ± 0.413</i> N=15; d=81	Identity	Intradaily variability distribution by site. Exact numerical summaries are in the adjacent cells. 
Exposure history													
meLEDI dose Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) <i>10.358 ± 16.216</i> N=141; d=761	9.296 (3.892, 21.895) <i>19.923 ± 27.784</i> N=13; d=71	10.495 (3.855, 17.395) <i>15.881 ± 20.395</i> N=13; d=74	6.751 (2.469, 14.435) <i>13.809 ± 23.711</i> N=18; d=97	3.862 (1.411, 10.217) <i>7.719 ± 9.847</i> N=26; d=137	7.11 (1.897, 15.789) <i>10.055 ± 11.237</i> N=10; d=55	3.317 (1.703, 6.29) <i>6.115 ± 8.79</i> N=23; d=125	4.35 (1.791, 10.625) <i>7.49 ± 8.898</i> N=17; d=96	2.195 (1.339, 4.117) <i>3.281 ± 2.573</i> N=6; d=30	6.045 (2.49, 14.907) <i>10.016 ± 10.799</i> N=15; d=76	Symlog (base 10; threshold 1)	meLEDI dose distribution by site. Exact numerical summaries are in the adjacent cells. 
Level													
Mean meLEDI Geometric average of daily meLEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	7.569 (4.55, 11.613) <i>9.171 ± 7.111</i> N=13; d=78	6.659 (4.18, 11.674) <i>8.274 ± 6.558</i> N=13; d=78	7.704 (3.081, 13.924) <i>12.017 ± 18.865</i> N=18; d=107	5.232 (3.224, 8.188) <i>7.221 ± 6.33</i> N=26; d=150	9.013 (4.011, 15.089) <i>12.102 ± 12.013</i> N=10; d=60	3.505 (1.469, 6.211) <i>4.166 ± 3.272</i> N=23; d=129	5.857 (3.726, 7.986) <i>7.778 ± 6.946</i> N=17; d=101	4.325 (3.585, 6.77) <i>5.248 ± 2.85</i> N=6; d=32	2.144 (0.744, 4.042) <i>3.019 ± 2.896</i> N=15; d=81	Symlog (base 10; threshold 1)	Mean meLEDI distribution by site. Exact numerical summaries are in the adjacent cells. 
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) <i>214.866 ± 508.56</i> N=141; d=816	252.689 (104.514, 461.953) <i>399.302 ± 518.102</i> N=13; d=78	175.974 (78.625, 299.106) <i>262.206 ± 329.201</i> N=13; d=78	156.184 (67.783, 339.731) <i>422.54 ± 1,209.616</i> N=18; d=107	100.057 (33.255, 209.904) <i>179.372 ± 241.53</i> N=26; d=150	130.78 (54.044, 259.726) <i>201.947 ± 194.042</i> N=10; d=60	101.102 (24.355, 218.457) <i>136.312 ± 134.07</i> N=23; d=129	83.24 (53.17, 151.786) <i>120.172 ± 101.545</i> N=17; d=101	68.143 (45.331, 132.458) <i>96.735 ± 72.311</i> N=6; d=32	46.66 (9.242, 103.175) <i>82.489 ± 115.07</i> N=15; d=81	Symlog (base 10; threshold 1)	Brightest 10 h mean distribution by site. Exact numerical summaries are in the adjacent cells. 
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) <i>0.243 ± 0.513</i> N=141; d=816	0.08 (0.016, 0.191) <i>0.183 ± 0.299</i> N=13; d=78	0.1 (0.035, 0.23) <i>0.2 ± 0.332</i> N=13; d=78	0.14 (0.036, 0.282) <i>0.223 ± 0.299</i> N=18; d=107	0.149 (0.074, 0.309) <i>0.273 ± 0.465</i> N=26; d=150	0.325 (0.119, 0.71) <i>0.515 ± 0.653</i> N=10; d=60	0.022 (0, 0.08) <i>0.054 ± 0.075</i> N=23; d=129	0.152 (0.052, 0.481) <i>0.475 ± 0.882</i> N=17; d=101	0.133 (0.055, 0.309) <i>0.411 ± 1.044</i> N=6; d=32	0.01 (0, 0.092) <i>0.059 ± 0.104</i> N=15; d=81	Symlog (base 10; threshold 1)	Darkest 10 h mean distribution by site. Exact numerical summaries are in the adjacent cells. 
Spectrum													

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) 0.724 ± 0.116 N=137; d=687	0.788 (0.725, 0.872) 0.796 ± 0.162 N=13; d=74	0.743 (0.699, 0.796) 0.743 ± 0.077 N=13; d=65	0.781 (0.675, 0.853) 0.768 ± 0.118 N=18; d=97	0.65 (0.592, 0.728) 0.667 ± 0.098 N=26; d=144	0.724 (0.652, 0.773) 0.72 ± 0.093 N=10; d=55	0.668 (0.629, 0.721) 0.673 ± 0.07 N=22; d=83	0.747 (0.634, 0.834) 0.735 ± 0.122 N=17; d=91	0.719 (0.661, 0.761) 0.706 ± 0.067 N=6; d=30	0.755 (0.729, 0.82) 0.758 ± 0.097 N=12; d=48	Identity	Melanopic daylight efficacy ratio distribution by site. Exact numerical summaries are in the adjacent cells. 
Timing													
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) 13:56 ± 01:51 N=141; d=816	13:16 (12:28, 14:10) 13:20 ± 01:33 N=13; d=78	14:12 (13:23, 15:18) 14:22 ± 01:51 N=13; d=78	13:52 (12:54, 15:04) 14:00 ± 01:42 N=18; d=107	13:38 (12:47, 14:44) 13:52 ± 01:42 N=26; d=150	14:54 (12:50, 16:03) 14:31 ± 02:14 N=10; d=60	13:44 (13:06, 15:20) 14:10 ± 01:51 N=23; d=129	14:10 (13:32, 15:21) 14:25 ± 01:37 N=17; d=101	12:34 (11:49, 14:02) 12:55 ± 01:44 N=6; d=32	12:49 (12:06, 13:39) 13:03 ± 01:58 N=15; d=81	Circular clock	Midpoint of the brightest 10 hours distribution by site. Exact numerical summaries are in the adjacent cells. 
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) 02:58 ± 01:41 N=141; d=816	02:02 (01:25, 02:48) 02:03 ± 01:07 N=13; d=78	02:55 (02:01, 03:27) 02:50 ± 02:05 N=13; d=78	02:32 (01:45, 03:22) 02:37 ± 01:19 N=18; d=107	03:08 (02:11, 04:10) 03:16 ± 01:34 N=26; d=150	02:37 (01:54, 04:17) 03:10 ± 02:00 N=10; d=60	03:39 (02:43, 04:34) 03:34 ± 01:49 N=23; d=129	03:31 (02:33, 04:14) 03:26 ± 01:25 N=17; d=101	01:45 (01:05, 03:01) 02:10 ± 01:27 N=6; d=32	02:33 (02:04, 03:17) 02:36 ± 01:36 N=15; d=81	Circular clock	Midpoint of the darkest 10 hours distribution by site. Exact numerical summaries are in the adjacent cells. 
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) 09:26 ± 02:22 N=140; d=727	07:57 (06:56, 08:33) 08:03 ± 01:32 N=13; d=74	09:04 (08:18, 09:52) 09:17 ± 01:56 N=13; d=68	08:36 (07:10, 10:13) 08:45 ± 02:21 N=18; d=93	09:43 (08:13, 10:53) 09:45 ± 02:21 N=25; d=133	09:36 (07:58, 10:53) 09:39 ± 03:12 N=10; d=56	09:32 (08:40, 11:21) 10:05 ± 02:18 N=23; d=118	10:01 (09:06, 11:52) 10:26 ± 02:28 N=17; d=95	08:11 (07:26, 09:57) 08:36 ± 01:30 N=6; d=29	08:43 (08:01, 10:10) 09:19 ± 01:52 N=15; d=61	Circular clock	First light timing above 250 lx melEDI distribution by site. Exact numerical summaries are in the adjacent cells. 
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) 18:03 ± 02:36 N=141; d=687	18:41 (17:38, 19:38) 18:39 ± 01:38 N=13; d=73	18:30 (17:37, 19:59) 18:37 ± 02:08 N=13; d=62	19:34 (18:04, 20:42) 19:16 ± 02:28 N=18; d=89	17:37 (15:35, 19:09) 17:18 ± 02:44 N=26; d=121	19:50 (18:37, 20:50) 19:33 ± 02:11 N=10; d=53	17:48 (16:46, 19:19) 18:01 ± 02:21 N=23; d=106	18:45 (17:02, 19:38) 18:23 ± 02:15 N=17; d=89	15:45 (13:20, 16:38) 15:22 ± 02:38 N=6; d=28	15:54 (14:34, 17:25) 15:46 ± 02:13 N=15; d=66	Circular clock	Last light timing above 250 lx melEDI distribution by site. Exact numerical summaries are in the adjacent cells. 

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Mean timing of exposure above 250 lx meEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) <i>13:31 ± 01:52</i> N=141; d=742	13:20 (12:29, 14:24) <i>13:21 ± 01:25</i> N=13; d=72	13:52 (13:13, 15:06) <i>14:02 ± 01:41</i> N=13; d=71	13:44 (12:28, 14:55) <i>13:51 ± 01:56</i> N=18; d=93	13:17 (12:17, 14:31) <i>13:24 ± 01:45</i> N=26; d=138	14:04 (12:58, 15:25) <i>14:09 ± 02:22</i> N=10; d=55	13:43 (12:55, 14:29) <i>13:47 ± 01:33</i> N=23; d=117	13:58 (12:54, 14:58) <i>14:04 ± 01:39</i> N=17; d=94	11:30 (10:28, 12:38) <i>11:26 ± 01:36</i> N=6; d=31	12:17 (11:04, 13:04) <i>12:11 ± 01:36</i> N=15; d=71	Circular clock	Mean timing of exposure above 250 lx meEDI distribution by site. Exact numerical summaries are in the adjacent cells.

Median (25th percentile, 75th percentile), *circular or arithmetic mean ± standard deviation*, N=participants; d=participant-days with a finite value for that metric.

¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.

² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.

³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.

Supplementary Table S2. Definitions, exact support, median (interquartile range) summaries, and density displays for the 17 near-eye personal light-exposure metrics overall and by site. Values use three decimal places where applicable. Metrics are distinct summaries and should not be interpreted as interchangeable outcomes.